LALIGENCE

LALIGENCE eKYC Project Proposal

Secure document verification, liveness, face matching, and face-based payment access

Prepared for	Prepared by	Date	Status
Customer / Internal Review	LALIGENCE eKYC Team	2026-05-21	Proposal draft

Proposal objective	Build and validate an IAL2-aligned eKYC workflow that verifies identity documents, checks fraud signals, confirms live user presence, matches the selfie to the document portrait, and enables returning face login or Face Pay authorization after enrollment.

1. Executive Summary

LALIGENCE is a web-first eKYC and secure access platform for identity verification and face-based authorization. The proposed system guides a user through document proofing, active liveness, hand gesture challenge, selfie capture, passive liveness, face matching, and final decisioning. After successful verification, the verified profile can be bound to one active Face ID for future login or payment authorization.

The current target is NIST IAL2-aligned readiness rather than a claim of formal certification.

2. Project Goals

- Provide a guided verification flow for passport and Lao ID card evidence.
- Detect edited portraits, edited dates, visual/MRZ mismatch, non-document upload, and poor-quality captures.
- Use active liveness and hand gestures to reduce static photo and replay attacks.
- Use passive liveness and face matching to compare selfie with document portrait.
- Store only verification results, verified profile attributes, and one active face template per user.
- Support returning face login and Face Pay transfer authorization for verified users.

3. Proposed Solution

The solution contains a React web app and Python backend. The frontend manages capture, user guidance, step gating, and status display. The backend performs document analysis, OCR/MRZ extraction, fraud scoring, liveness analysis, face matching, profile storage, and face-login matching.

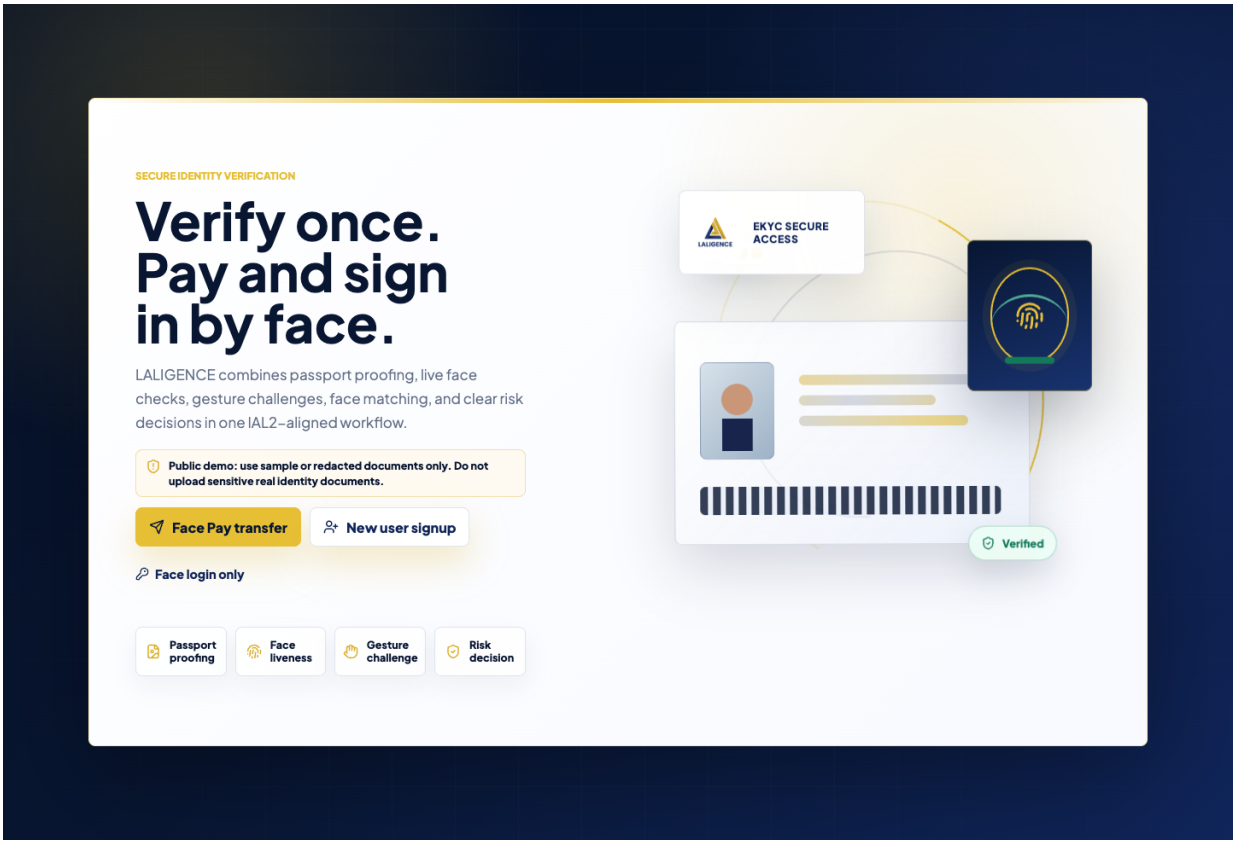
Layer	Role	Key functions
Frontend	User experience and capture	Document upload/camera, liveness, gesture UI, selfie, face login, payment authorization
Backend	Verification intelligence	OCR, fraud scoring, face matching, anti-spoof checks, profile/session APIs
Database	Verification and profile records	One user ID to one active Face ID; one passport number to one verified profile
Models	Local/open-source AI	Face detection/recognition, anti-spoofing, OCR/MRZ, document fraud checks

4. End-to-End Workflow

1. New user starts signup and receives or enters a user ID.
2. User uploads or captures passport/Lao ID evidence.
3. System checks document quality, OCR/MRZ, expiry, and fraud signals.
4. User performs active face liveness challenges.
5. User completes hand gesture challenges inside a moving target circle.
6. User captures selfie for passive liveness and face matching.
7. System returns final decision, scores, reason codes, and OCR summary.
8. Accepted user can enroll Face ID for future login and Face Pay authorization.

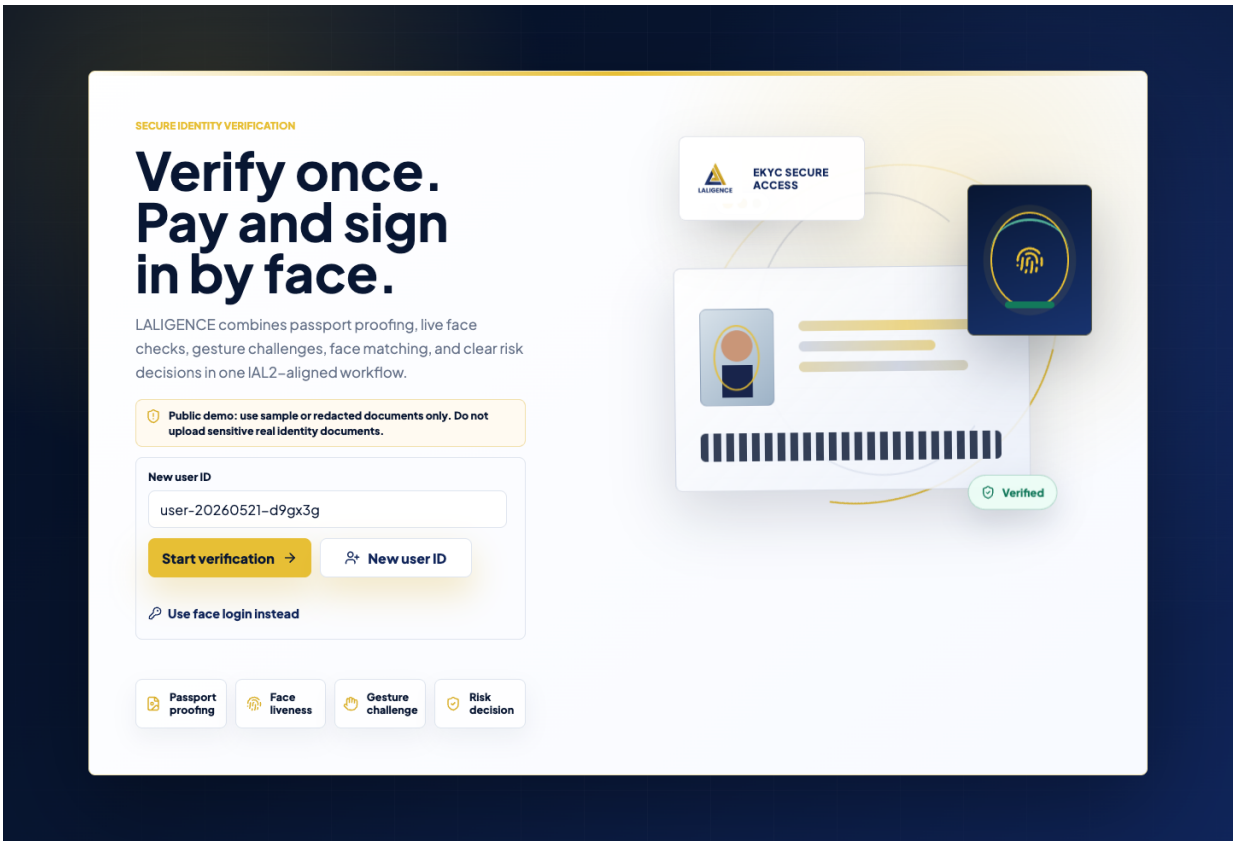
5. Usage Guide With Screenshots

Start Page



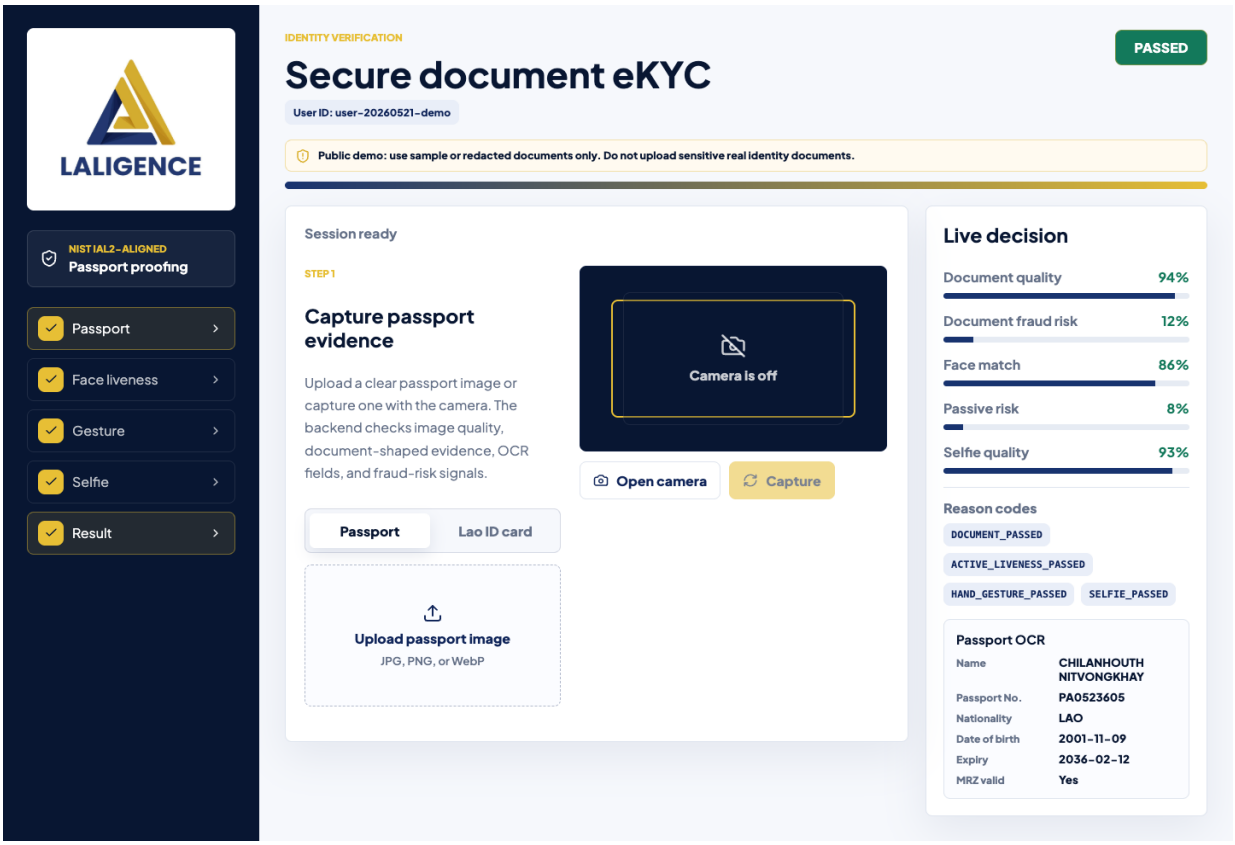
Landing screen with Face Pay transfer, new user signup, face login, and capability cards.

New User Signup



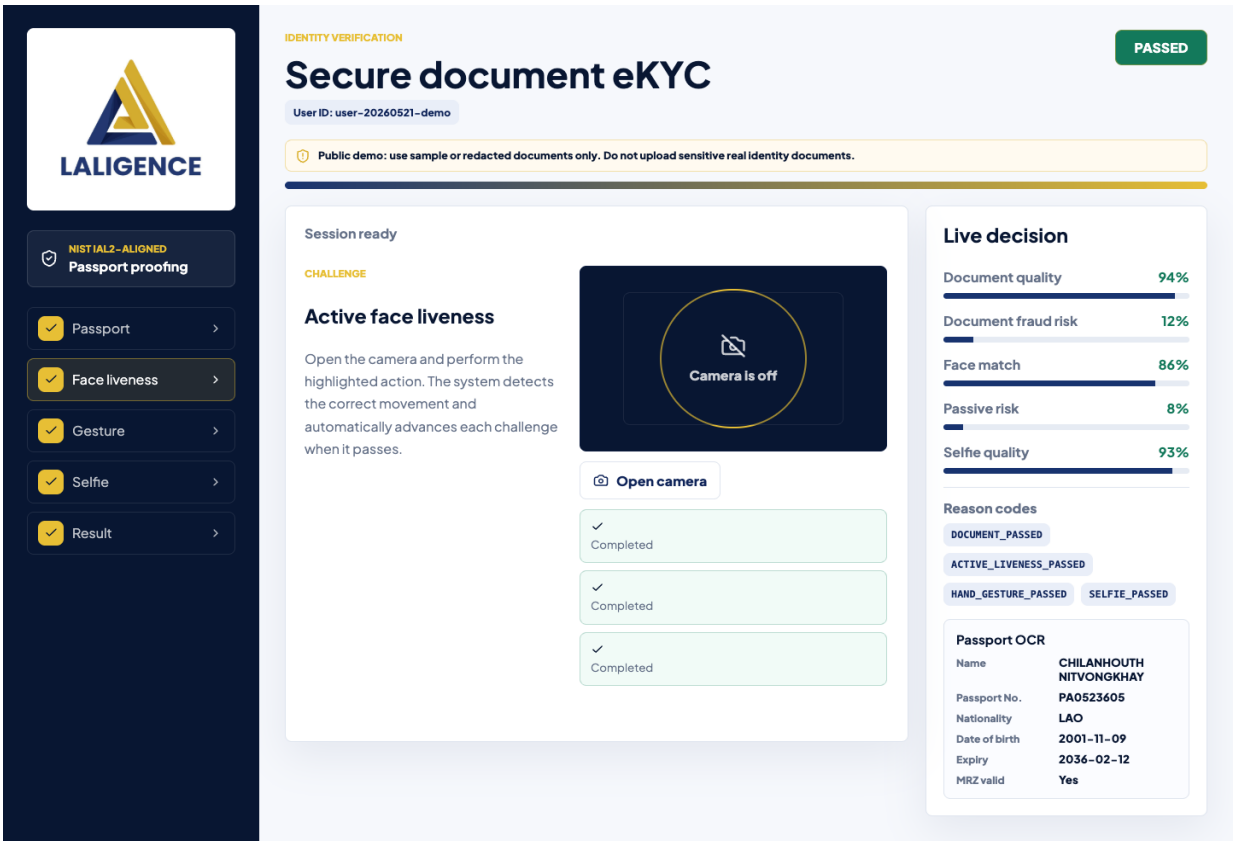
New users start verification with a generated or entered user ID.

Document Proofing



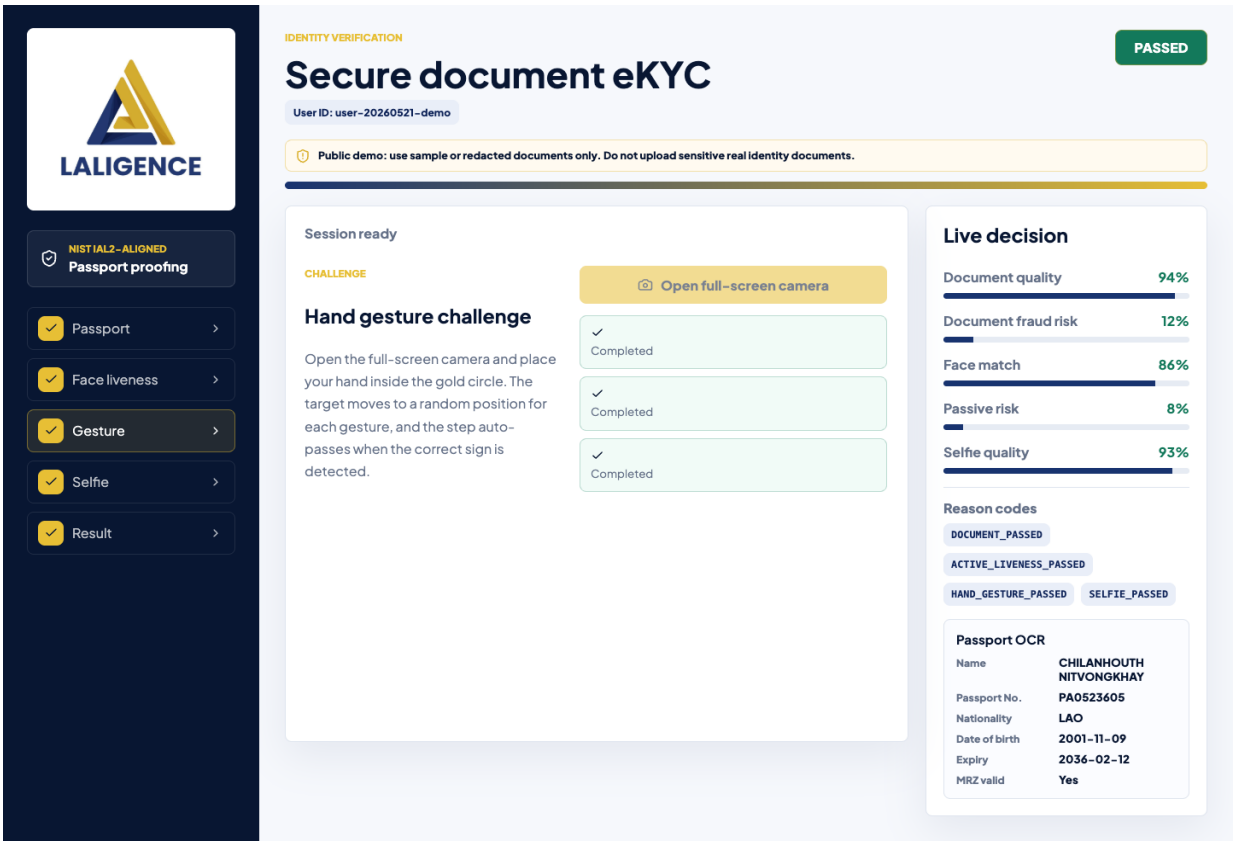
The user uploads or captures passport/Lao ID evidence and sees document quality, fraud risk, and OCR results.

Active Face Liveness



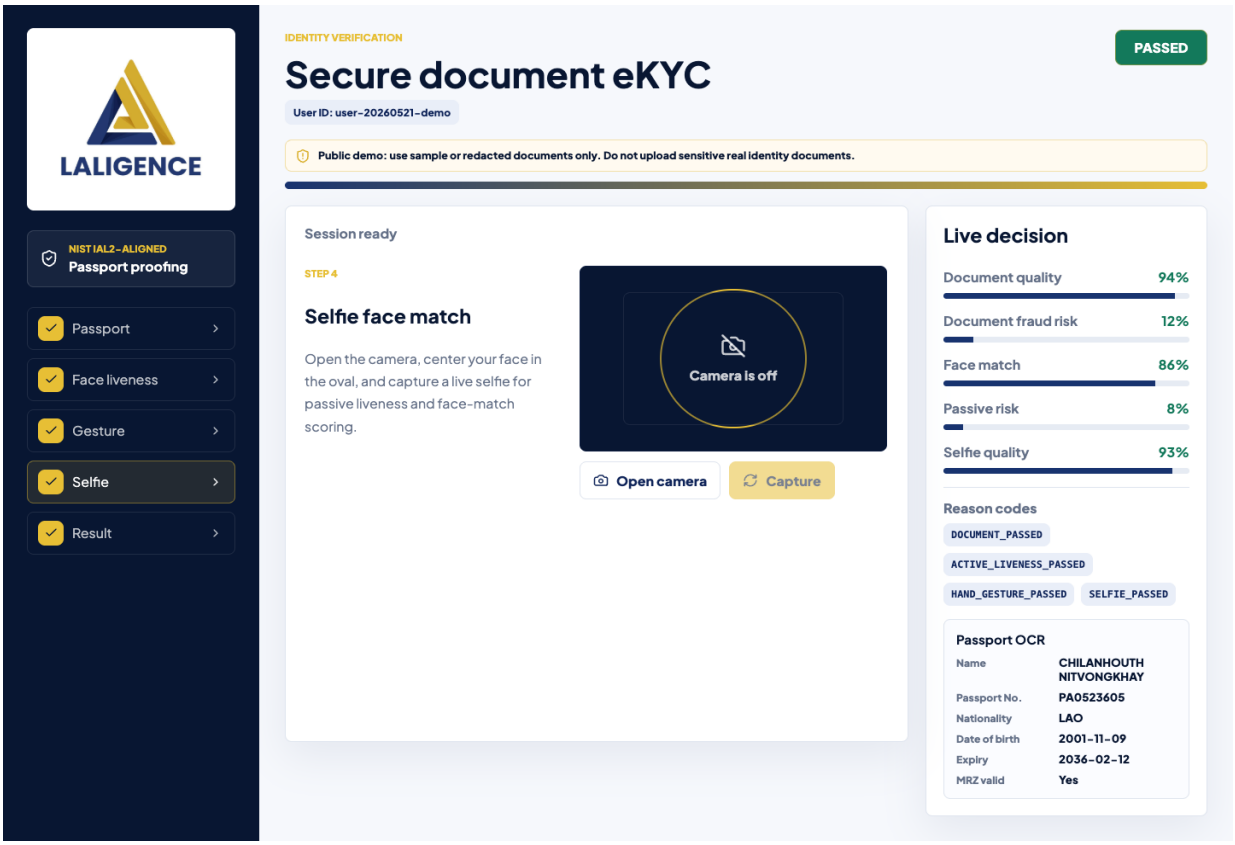
The user performs live camera actions such as blink, head movement, and mouth opening.

Hand Gesture Challenge



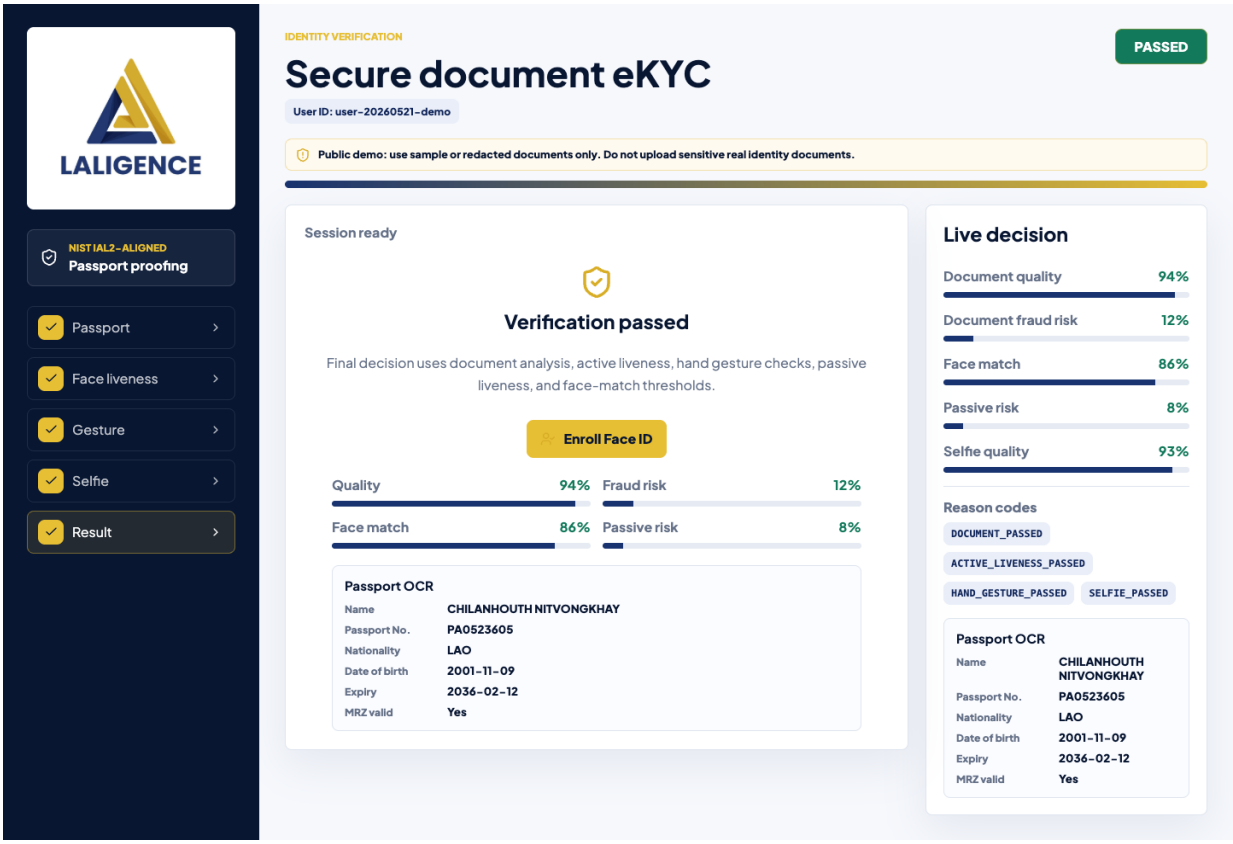
The user performs requested hand gestures in a moving target area.

Selfie Face Match



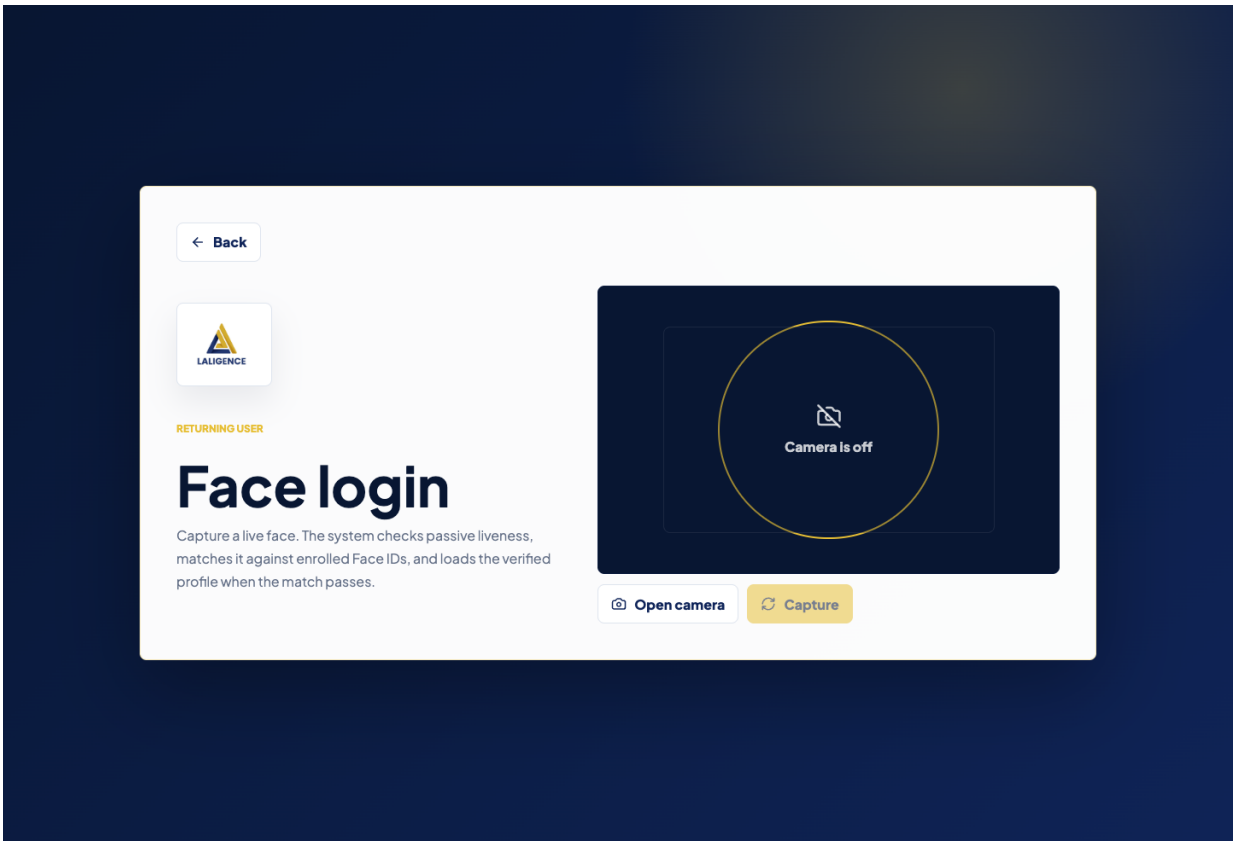
The user captures a selfie for passive liveness and face matching.

Result Decision



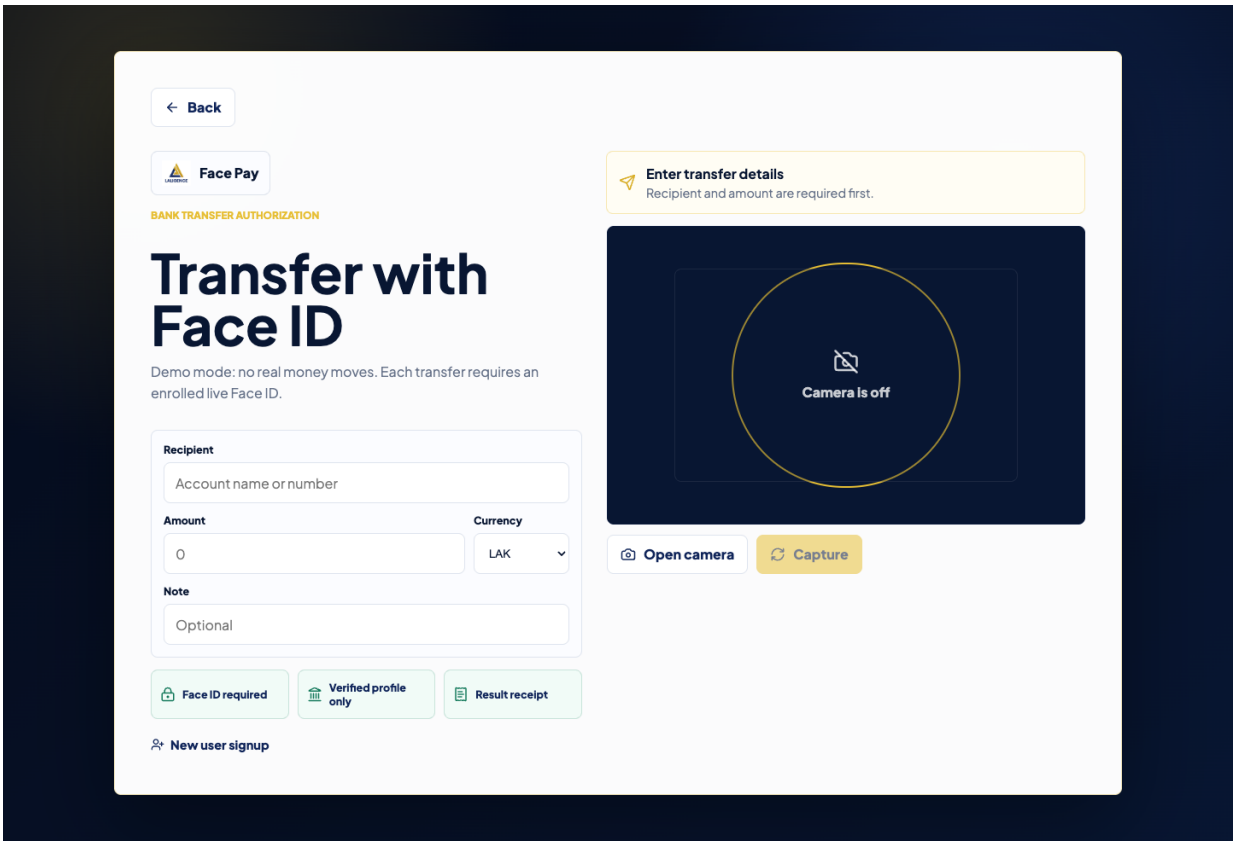
The final result shows decision, OCR, biometric scores, reason codes, and Face ID enrollment.

Face Login



Returning verified users can use live face login instead of repeating full onboarding.

Face Pay Transfer



Payment-style authorization requires verified face scan before transfer approval.

6. Security and Fraud Detection Scope

Control area	Examples covered	Expected behavior
Document fraud	Edited portrait, edited expiry, text overlay, fake template, recaptured document	Reject or route to manual review with reason codes
Document validity	Expiry, MRZ checksum, OCR/MRZ consistency, document number	Reject invalid, expired, or inconsistent evidence
Active liveness	Blink, mouth open, head turn, screen replay attempts	Pass only when correct live human action is detected
Gesture challenge	Random moving target with simple hand signs	Pass only when correct gesture is detected inside target
Passive liveness	Paper photo, phone screen, laptop screen, multiple faces	Reject spoof or replay risk
Face matching	Same person vs impostor selfie	Accept above threshold; reject low-match or multiple-face cases

7. NIST IAL2 Alignment

The app is intended to support NIST SP 800-63A-4 IAL2-aligned identity proofing. It supports evidence collection, attribute extraction, evidence validation, biometric ownership verification, liveness checks, workflow gating, and audit-style reason codes.

Compliance note	A formal IAL2 claim requires source validation, privacy notices, retention, deletion, audit logging, notification of proofing, operating procedures, and security controls.

8. Implementation Plan

Phase	Deliverables	Success criteria
1. Web pilot	Frontend and backend demo, document proofing, live selfie, result page	Core web end-to-end tests
2. Model tuning	Dataset tests for document fraud, face match, passive liveness, active liveness	Measured FAR/FRR and tuned thresholds
3. Profile and payment	Face ID enrollment, face login, Face Pay transfer authorization	One active Face ID per user and payment scan works
4. Compliance readiness	IAL2 test evidence, audit logs, privacy/retention docs, security controls	Review package prepared for customer review
5. Production hardening	Cloud deployment, database, monitoring, rate limiting, backup	Stable multi-user testing with acceptable latency

9. Recommendation

Proceed with a controlled pilot for internal users and customer demonstration. Use the prepared NIST IAL2 test matrix to measure accuracy, spoof rejection, document fraud rejection, and usability before expanding to production or regulated payment use cases.